

Introducing the Super OSSM

A clean sheet, modular design, for a choose your OSSM build adventure.

Building on community improvements to the R+D design, the Super OSSM is compatible with 4 motor types, 2 PCBs, 3 rail & belt sizes, with 2 drive types to choose from, plus add-on modules.

My approach is to make it compact and compatible, only increasing the diameter by 15mm from the standard PitClamp ring, whilst being able to offer a wide range of compatibility for different sized components.



Large focus on - reducing the number of parts required, with the main unit being only 4 3D printed parts - simplifying the BOM, with standardised bolt sizes & lengths and using standard OSSM parts - ease of 3D printing, by minimising the need for support material - and improved assembly & serviceability, with a twist-lock to access the motor and PCB.

The final goal has been to improve the overall aesthetics so that the Super OSSM looks like a finished product rather than a DIY project.

Unfortunately, two design compromises were made - first, support and compatibility for the legacy iHSV motor was not possible, due to the backpack electronics that it has, making packaging inside the Super OSSM impossible - second, the use of 688 (8x16x8mm) bearings, rather than standard OSSM bearings, to aid ease of assembly.

Design features include a larger version of @Armpit's PitClamp design with 120 position / 3° increments, and a fixed or folding clamp option for easy under-bed storage.

Support for @BuildingRandomStuff's ALT PCB and the R+D PCB, with a translucent logo for the status LED.

An alternative drive option to use @SaladDressing's Capstan drive system.

Support for -

Motors	: 42AIM, 57AIM, 60AIM series & 60AIM40F	
PCB	: R+D & ALT	
Rails	: MGN12, MGN15 & HGR20	
Drive Mechanism	: Belt or Capstan	
Belts	: 2MGT, 3MGT & HTD-3M	
Add-ons	: Stroker Module	(hopefully more in future)

Special thanks to @Frayd for their input and putting up with me being "extra" on calls to discuss the project.

I hope to have designed an OSSM for one and an OSSM for all.

This build and assembly guide is written to give an overview of the 3D printing, build and assembly process for the Super OSSM, from printing to play.

Print order and assembly steps are set out in a logical order for you to start building your Super OSSM as soon as the first parts are 3D printed. Due to the modularity of the design, not all configurations may be shown, and common sense may be needed. If bolt lengths are not specified in a step, please refer to the BOM.

First time builders it is recommended that you join the Kinky Makers Discord server  as it is a wealth of knowledge, a great place to chat with like-minded Kinky Makers, share and show-off projects, get help and technical or troubleshooting advice, and contribute to the Open-Source Sex Machine project.

Step 1 : Choose your OSSM build adventure

- Determine if you want a Standard, Lite or XL build - depending on the size of toys and attachments you plan to use, including any add-ons.
- Select an appropriate motor and rail size.
- Decide on drive system - belt or capstan.
- Choose your PCB.

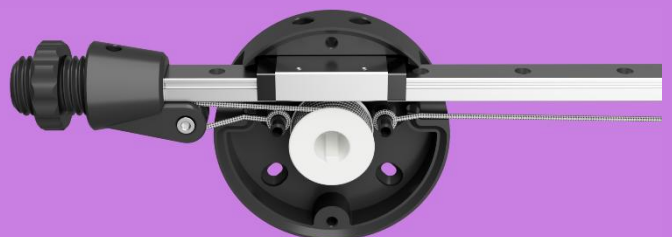


Recommended Configurations

Build	Head	PCB	Motor	Drive Mechanism	Belt	Rail Type	Rail Length	Attachment Weight	Add-ons*
Lite	Std	R+D ALT	42AIM 57AIM	Belt	2MGT x 12 mm	MGN12	≤ 400 mm	≤ 2 kg	Stroker Module
Std	Std	R+D ALT	57AIM	Belt Capstan	2MGT x 15 mm 3MGT x 15 mm 1.5~2 mm Dyneema	MGN15	350 ~ 500 mm	2 ~ 3kg	Stroker Module
XL	XL	R+D ALT	60AIM 60AIM40F	Belt Capstan	HTD-3M x 15 mm 1.5~2 mm Dyneema	HG20R	≥ 500 mm	≥ 3 kg	Stroker Module

* High-torque motors are required to operate the Super OSSM as a stroker and poker simultaneously.

Parts are modular so you can customise your Super OSSM, depending on preferences and parts availability.



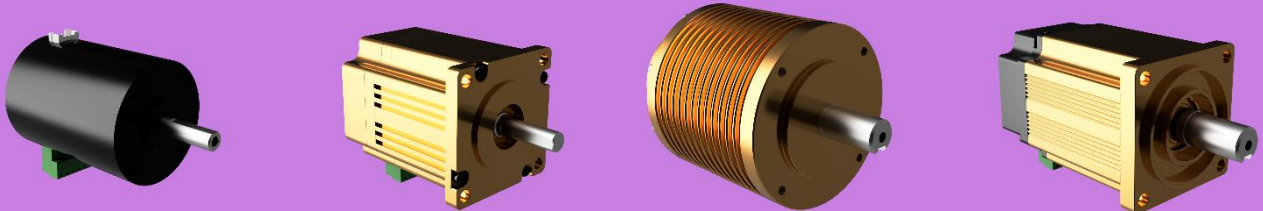
Compatible Motors

Motor	Protocol**	Dimensions	Shaft Size	Rated Speed	Max Speed	Torque	Voltage	Current	Power
42AIM10	RS485	∅ 53 x 42 mm	∅ 8 x 23 mm	1000 RPM	1500 RPM	0.33 NM	24 ~ 36 V	1.5 A	33 W
42AIM15	RS485	∅ 53 x 55 mm	∅ 8 x 27 mm	1000 RPM	1500 RPM	0.48 NM	24 ~ 36 V	2.2 A	50 W
42AIM30*	RS485	∅ 53 x 70 mm	∅ 8 x 27 mm	1000 RPM	1500 RPM	0.96 NM	24 ~ 36 V	4.4 A	100 W
57AIM15	RS485	57 x 57 x 57 mm	∅ 8 x 26 mm	1000 RPM	1500 RPM	0.48 NM	24 ~ 36 V	2.2 A	50 W
57AIM30*	RS485	57 x 57 x 72 mm	∅ 8 x 26 mm	1000 RPM	1500 RPM	0.96 NM	24 ~ 36 V	4.4 A	100 W
60AIM25*	RS485	∅ 80 x 76 mm	∅ 14 x 30 mm	1000 RPM	1500 RPM	2 NM	36 V	7.0 A	200 W
60AIM40*	RS485	∅ 80 x 90 mm	∅ 14 x 30 mm	1000 RPM	1500 RPM	3 NM	36 V	10.0 A	350 W
60AIM40F*	RS485	60 x 60 x 92 mm	∅ 14 x 30 mm	1000 RPM	1500 RPM	2 NM	36 V	7.0 A	200 W

* Recommended motors ** Recognisable by the **GREEN** terminal on the motor

Note : 57AIM30 motor is often referred to as the “Gold” motor, and is the standard OSSM motor available from R+D.

Comparison video of some of the available motors



Compatible Rails

Rail	Dimensions	Hole Size	Hole Spacing	Length	Carriage	Dimensions	Bolt Size	Bolt Spacing
MGN12	12 x 8 mm	M3	25 mm	≤ 400 mm	MGN12H	45 x 27 x 13 mm	M3 x 3.5 mm	20 x 20 mm
MGN15	15 x 10 mm	M3	40 mm	350 ~ 500 mm	MGN15H	59 x 32 x 16 mm	M3 x 4 mm	25 x 25 mm
HGR20	20 x 17.5 mm	M5	60 mm	≥ 500 mm	HGH20HA QHH20A	92 x 44 x 30 mm	M5 x 6 mm	50 x 32 mm

Max Stroke Length = Rail Length – Head Width – Rail Ends

Head Width : Std 70 mm | XL 95 mm

Rail Ends : MGN12 65 mm | MGN15 80 mm | HGR20 80 mm

Rail Length [400 mm] – Std Head Width [70 mm] – MGN15 Rail Ends [80 mm] = Max Stroke Length [250 mm]



Step 20 : Order parts

- See the BOM for the full parts list. 
- You can order a kit from R+D - including 57AIM “Gold” motor, PCB + Remote, Rail & Hardware
- or source parts yourself (required for alternative builds)

Step 3 : Calibrate your 3D printer



Recommended print settings :

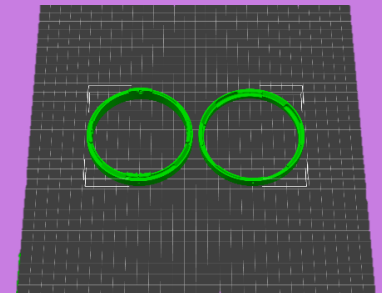
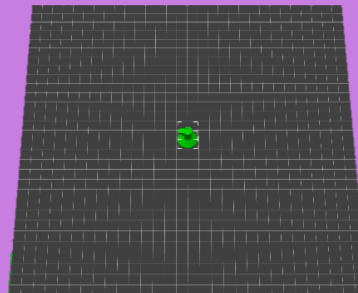
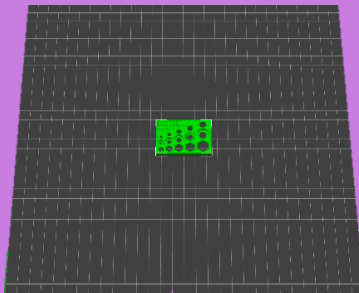
- Material - PETG, ABS, PLA+
- Nozzle - 0.4 mm
- Perimeters - 4 | 8 for threaded rail ends (minimum for strength vs print time)
- Infill - 25% | 50% for threaded rail ends
- Layer height - 0.16 mm
- Wall order - Inside / Outside (improves quality of threads and overhangs)
- Support - NO for test prints | 40° for parts (recommend paint-on supports)

⚠ DO NOT USE CF FILAMENTS! ⚠

Note : It is recommended that all parts for the project are printed on the same 3D printer to ensure that all parts fit together and account for any minor variations.

Print the included calibration files to confirm dimensional accuracy :

- Check that the test print is the correct size 50 x 35 x 5 mm (± 0.1 mm) and the fasteners fit through the holes.
 - Top holes are actual size - M3 = 3.0 mm
 - Middle holes are 5% larger - M3 = 3.15 mm
 - Counterbores are 0.5 mm larger than the fastener head - M3 head 5.5 mm = 6.0 mm
 - Hexes are actual size for press fit so the captive nuts are securely - M3 hex 5.5 mm = 5.5 mm
- Print-in-place button has a 0.3 mm offset for a snug fit, which does not rattle, whilst being easy to print and operate smoothly. The button should require a firm push to free it after printing.
- Twist-lock should be a tight fit, to account for wear from use, and will operate more freely with a satisfying click after opening and closing a couple of times.



Troubleshooting :

If you have difficulty with the test prints, check the following settings BEFORE you print the Super OSSM.

- Extrusion multiplier - complete a Flow Ratio test to ensure that your printer is not over-extruding as over-extrusion can cause the print-in-place button to print solid, undersized holes and parts that do not fit together. (OrcaSlicer has built-in calibration tools)
- XY contour compensation - if parts are over or undersized and do not fit together.
- XY hole compensation - if holes are undersized whilst the rest of the part is the correct dimension [0.15 mm recommended] (if your preferred slicer has the setting)
- Dynamic speeds and cooling for bridging and overhangs - there are parts that need clean 45° surfaces and bridged overhangs for embossed text.
- General print settings, including supports for best print quality and easy removal - whilst the Super OSSM has been designed to minimise support material where possible, there are areas where it is unavoidable such as spaces for captive nuts, PCB ports etc.

Step 4 : Get ready

Clear a workspace and organise your tools, you will need the following:

- Allen key set - 1.5 mm ~ 5 mm
- Small flat screwdriver
- Long-nose pliers
- Hook & pick set (recommended to help remove support material)

Step 5 : Head (Base)

- Print the Base of the head - Std or XL depending on your build.
- Insert (4) M4 captive Nylon locknuts into the hex holes on the back side of the head.
- Insert (2) M4 bolts through the bearing carriers. (recommended to prevent accidental breakage)
- Insert (2) M3 captive nuts into the slot at the top of the head. (Capstan drive only)

Step 6 : Motor

- Setup your motor by flashing it with the Web Flasher tool (See Motor setup guide)

Step 7 : Motor Ring

- Print the appropriate Motor Ring for your motor.
- Insert (4) M5 captive locknuts into the slots inside the motor ring
- Slide the motor into the motor ring, from the head side.
- The motor will be secured with the bolts used to attach the head in the next step.
- If you have chosen either the 42AIM or 60AIM (round) motors, see additional steps 6A & 6B.

Step 7A : 42AIM

- Insert the motor into the motor ring from the PCB side and secure with (4) M3 x 6 mm countersunk bolts.

Step 7B : 60AIM

- DO NOT insert the (4) M5 Captive nuts into motor ring.
- Insert the motor into the motor ring from the PCB side and line up the mounting holes.

Step 8 : Attach the Head

- Place the Head (Base) over the motor shaft, aligning the mounting holes.
- Attach the Head to the motor and ring assembly with (4) M5 countersunk bolts.

Step 9 : Pulley

- Put the pulley on the motor shaft.
- Loosely fasten the grubscrews, as the pulley will be aligned with the belt later.
- If you are using a Capstan drive, you can secure the pulley to the shaft with the countersunk bolt.

Step 10 : Bearings

(Not required for Capstan drive)

- Place (4) 688 bearings over each bearing carrier.

Step 11 : Rail

- Add the rail linear bearing block to the Head.
- You will need to use the spacer for the MGN12 rail.
- Secure the linear bearing with (2) M3 or M5 bolts.

Step 12 : Belt Drive

Step 12A: Tensioner

- Print the appropriate size Tensioner and Keystone for your rail size and belt type.
- Print Tensioner cap.
- Insert (1) M4 nut into the slot in the tensioner through the large opening for the rail.
- Insert (1) M4 bolt from the opposite side to hold the nut in place.
- Only screw the bolt in until the nut is flush with the end of the bolt, as this nut and bolt is used to tighten the belt.
- Feed the belt through the slot below the rail opening.
- Place the keystone on the belt, leaving a small tail (2 ~ 3 cm).
- Pull the belt back through the slot until the Keystone is wedged firmly in place.
- Push the Tensioner end on to the rail, until it is all the way in the slot.
- Screw the Tensioner cap on the end, tucking the tail of the belt inside.

Step 12B: Belt

- Feed the belt over the bearings and under the pulley.
- Gently pull tight to remove any slack.
- Cut the belt to length, it should end just before the last hole on the rail.

Step 12C: Fixed End

- Print the appropriate size Fixed End and Belt Clamp for your rail size and belt type.
- Put the Belt Clamp between the belt and rail , aligning the hole with the last hole of the rail.
- Slide the Fixed End over the Rail, Belt and Belt Clamp.
- Secure in place with (1) M4 or M5 locknut and (1) M4 or M5 bolt.
- The head of the bolt will sit in the counterbore of the rail and pull the Fixed End tight against the underside of the rail, sandwiching the Belt and Belt Clamp in place.

Step 12D: Align the Pulley

- Gently move the rail through the full stroke length checking the belt tracking and pulley alignment.
- Adjust the pulley so that the belt does not touch either side through the full range.
- Secure the pulley to the motor shaft by tightening the grubscrews.



Step 13 : Capstan Drive

Step 13A: Rail Ends

- Print appropriate size (2) Capstan Rail Ends and (1) Cap
- Insert (1) M4 nut into the slot in the rail end through the large opening for the rail.
- Insert (1) M4 bolt from the opposite side to hold the nut in place.
- Only screw the bolt in until the nut is flush with the end of the bolt, as this nut and bolt is used to tighten the belt.
- Insert (2) U604-ZZ Deep Groove bearings into the slot.
- Secure in place using (2) M4 x 25 mm bolts and (2) M4 Nylon locknuts.
- Push the Rail Ends on to either side of the rail, until it is all the way in the slot.

Step 13B: Capstan Rope

- Place the Dyneema rope in the RIGHT slot at the top of the head.
- Wrap the end of the Dyneema around the post at the top of the head.
- Secure the end of the Dyneema in place with (1) M3 bolt and (1) M3 flat washer.
- Feed the opposite end of the Dyneema through the (2) idler pulleys in the RIGHT rail end from the top.
- Run the Dyneema along the underside of the rail to the pulley.
- Wrap the Dyneema 8 times around the pulley starting from the back.
- Run the Dyneema along the underside of the rail to the opposite rail end.
- Feed the running end of the Dyneema through the (2) idler pulleys in the LEFT rail end from the bottom.
- Place the Dyneema rope in the LEFT slot at the top of the head.
- Wrap the end of the Dyneema around the post at the top of the head, pulling tight to remove ALL slack from the rope.
- Secure the end of the Dyneema in place with (1) M3 bolt and (1) M3 flat washer.

Step 14 : Cover

- Print the Cover for the head - Std or XL depending on your build.
- Remove the (2) M4 bolts in the bearing carriers.
- Place the cover over the head assembly.
- Secure in place with (4) M4 bolts on the face of the Cover, and (2) M3 or M5 bolts at the top into the Linear Bearing carriage.
- Make sure the cover is fully closed without a gap before you insert the bolts at the top, to prevent cross threading the threaded holes of Linear Bearing carriage.

Step 15 : Tension

- Tension the Belt or Dyneema rope using the bolt in Tensioner or both Capstan Rail ends.
- DO NOT TENSION UNLESS THE COVER IS BOLTED IN PLACE! - Tensioning without the Cover being bolted in place can cause damage to the bearing carriers and or mounting for the Linear Bearing.

Step 16 : PCB

- Flash OSSM firmware on to the PCB. (See Firmware setup guide)
- Attach the Motor wires to the PCB.

Step 17 : PCB Housing

- Print the PCB housing for your PCB type – R+D or ALT PCB
- Print the Diffuser using translucent filament and insert in place.
(Recommend 100% infill with Concentric Top, Bottom & Infill pattern, XY Contour -0.2 mm)
- Insert the M3 captive nuts.
- Insert the PCB into the PCB Housing, making sure the buttons and ports align correctly.
- Secure in place with M3 x 10 mm bolts.
- Insert (4) M2 captive nuts around the inside perimeter.
- Plug in Motor.
- Twist the PCB Housing assembly in place on the rear of the Motor Ring until it clicks and the dot lines up with the lock icon.
- Lock in place with (2) M2 screws. (Optional)

Step 18 : Clamp

- Print the parts of the Clamp
- Insert the M4 captive Nylon locknuts into their slots in the Base and Dogbone.
- Insert the M4 captive nut into the slot in the Handle.
- Using M4 x 25 mm bolts attach the Arm(s) to the base.
- Using (1) M4 x 12 mm bolt attach the Handle to the Locking arm.
- Using (1) M4 x 25 mm bolt and (2) M4 flat washers attach the Dogbones to the handle.
- Insert the spacer between the Dogbones and secure with (1) M4 x 25 mm bolt.
- Insert (2) M6 x 12 mm bolts into the holes in the base and attach (1) M6 Double T-nut or (2) M6 Single T-nuts.
- Attach the Clamp to a stand or mounting of your choice.

Step 19 : End Effectors

- Print and attach the End Effector of your choice to the M24 thread on Rail End(s)

Step 20 : Enjoy

- Connect and Control your Super OSSM with a Remote, Web Controller or App.
- Share your build in the Kinky Makers Discord.
- Enjoy using your Super OSSM!

Add-on : Stroker Module

Step A : Print

- Print the appropriate size Add-on parts for your Super OSSM head type, Cylinder size and length.

Step B : Stroker Base

- Release all the tension from the Belt or Capstan Rope, by backing off the tensioning bolt(s).
- Remove the (2) M4 bolts in the middle that go through the bearing carriers.
- Place the Stroker Base on the face of the Head.
- Secure in place with (2) M4 bolts 10 mm longer than the ones you removed.
- Re-tension the Belt or Capstan Rope.

Step C : Stroker Top

- Remove the (2) M4 bolts at the top and bottom of the head.
- Place the Cylinder on the Stroker Base.
- It is important that you print the correct length Stroker Base as it keeps the Stroker in position during use.
- Place the Stroker Top over the cylinder.
- Secure in place with (2) M4 bolts 10 mm longer than the ones you removed.

Step D : Stroker Cylinder

- Connect the Stroker Arm to the Cylinder with the Stroker Cap.
- Connect the Stroker Arm to the rail, you can use a Jam nut or End Effector.

Step E : Stroker Receiver

- Connect the Stroker Arm to the Cylinder with the Stroker Cap.
- Connect the Stroker Arm to the rail, you can use a Jam nut or End Effector.

Step F : Homing & Control

- Select Stroker Mode on your Remote, Web Controller or App.
- Home your Super OSSM with the Stroker Module attached.

Bill of Materials (BOM)

Qty	Part	Description	Notes
Electronics			
1	Motor	42AIM, 57AIM, 60AIM, 60AIM40F	See recommended builds for more information
1	PCB	R+D or ALT	
1	* Remote	Wired remote or RADR	Wired remote for R+D PCB only
Rails			
1	MGN12	≤ 400 mm	Std Head
1	MGN15	350 ~ 500 mm	Std Head
1	HGR20	≥ 500 mm	XL Head
<i>Select one, size / type depending on your Super OSSM build</i>			
1	MGN12H	MGN12 rail	Std Head
1	MGN15H	MGN15 rail	Std Head
1	HGH20HA / QHH20A	HGR20 rail	XL Head
<i>Select one, size / type depending on your Super OSSM build</i>			
4	M3 x 12 mm	Socket head	MGN12 rail
4	M3 x 10 mm	Socket head	MGN15 rail
4	M5 x 15 mm	Socket head	HGR20 rail
<i>Select one, size / type depending on your Super OSSM build</i>			
Belts & Pulleys			
1	2MGT x 12 mm	~1 m length, cut to size	MGN12 rail
1	3MGT x 12 mm	~1 m length, cut to size	MGN12 rail
1	2MGT x 15 mm	~1 m length, cut to size	MGN15 rail
1	3MGT x 15 mm	~1 m length, cut to size	MGN15 rail
1	HTD-3M x 15 mm	~1 m length, cut to size	HGR20 rail
1	Dyneema 1.5 mm	~2 m length, cut to size	Capstan Drive
<i>Select one, size / type depending on your Super OSSM build</i>			
1	2MGT 20T 8 mm Bore Pulley	12 ~ 15 mm for belt width	Motors - 42AIM, 57AIM
1	3MGT 20T 8 mm Bore Pulley	12 ~ 15 mm for belt width	Motors - 42AIM, 57AIM
1	HTD-3M 25T 14 mm Bore Pulley	15 mm for belt width	Motors - 60AIM, 60AIM40F
1	Capstan Pulley	Print version for motor type & head size	Motors - ALL
<i>Select one, size / type depending on your Super OSSM build</i>			
* Bearings			
8	688-ZZ / 688-2RS	8 x 16 x 5 mm	Required for Belt drive only
Head			
4	M4 x 30 mm	Socket head	Std Head
4	M4 x 50 mm	Socket head	XL Head
4	M4 nut	Nylon locknut	
Motor Ring			
4	* M3 x 6 mm	Countersunk head	Motor Ring - 42AIM
4	M5 x 15 mm	Countersunk head	Std Head Motor Ring - 60AIM
4	M5 x 20 mm	Countersunk head	Std Head Motor Ring - 42AIM, 57AIM, 60AIM40F XL Head Motor Ring - 60AIM
4	M5 x 25 mm	Countersunk head	XL Head Motor Ring - 42 AIM, 57AIM, 60AIM40F
4	* M5 nut	Nylon locknut	Motor Ring - 42 AIM, 57AIM, 60AIM40F
<i>Select, size / type depending on your Super OSSM build</i>			

Qty	Part	Description	Notes
PCB Housing			
4	* M2 nut	Standard nut	Optional to lock the PCB cover
2	* M2 x 6 mm	Button or Socket head	Optional to lock the PCB cover
#	M3 nut	Standard nut	(3) R+D PCB (2) ALT PCB
#	M3 x 10 mm	Button or Socket head	(3) R+D PCB (2) ALT PCB
* Rail Ends			
1	M4 nut	Standard nut	Tensioner
1	M4 x 25 mm	Socket head	Tensioner (MGN12 & MGN15)
1	M4 x 30 mm	Socket head	Tensioner (HGR20)
1	M3 nut	Nylon locknut	Fixed end (MGN12 & MGN15)
1	M3 x 18 mm	Socket head	Fixed end (MGN12 & MGN15)
1	M5 nut	Nylon locknut	Fixed end (HGR20)
1	M5 x 22 mm	Socket head	Fixed end (HGR20)
<i>Select, size / type depending on your Super OSSM build</i>			
* Capstan Drive			
2	M3 washer	Flat washer	Std Head - 7 mm OD XL Head - 10 mm OD
2	M3 nut	Standard nut	
2	M3 x 6 mm	Socket head	Std Head
2	M3 x 10 mm	Socket head	XL Head
4	U604ZZ	Groove bearing 4 x 13 x 4 mm	
6	M4 nut	Nylon locknut	
4	M4 x 25 mm	Socket head	Bearings
2	M4 x 50 mm	Socket head	Tensioner
1	M4 x ## mm	Countersunk head	Pulley to motor (42AIM & 57AIM)
1	M5 x ## mm	Countersunk head	Pulley to motor (60AIM & 60AIM40F)
Clamp			
2	M4 washer	Flat washer	Max OD 8 mm
1	M4 nut	Standard nut	
4	M4 nut	Nylon locknut	
1	M4 x 12 mm	Socket head	
4	M4 x 25 mm	Socket head	
2	M6 x 12 mm	Socket head	
1	M6 Double T-Nut	25 mm hole spacing	Can use (2) M6 Single T-nuts
#	Furniture Pad	38 mm / 1.5 inch self-adhesive	Desk and Bed mounts

* Optional part * Additional part

Note : Due to the modular design of the Super OSSM, parts for all build configurations listed in the BOM.
Please check the Notes and compile YOUR parts list carefully before ordering.